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Anomaly detection in multidimensional time series for water injection pump operations based on LSTMA-AE and mechanism constraints

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Addressing the issues of inadequate information exchange among subsequences in the operational time series of water injection pumps, leading to low accuracy and high false alarm rates in anomaly detection, this paper proposes a multidimensional time series anomaly detection method for water injection pump operations, leveraging Long Short-Term Memory Autoencoder augmented with Attention Mechanism (LSTMA-AE) and mechanistic constraints. The LSTMA-AE framework encompasses three primary modules: a Time Feature Extraction Module (Encoder), an Attention Layer, and a Data Reconstruction Module (Decoder). The Encoder captures temporal dependencies and features within the input sequences, mapping the input data into a higher-dimensional space. The Attention Layer, embedded within the hidden state computation, dynamically adjusts the contribution of each timestep's input information to the hidden state, thereby enhancing the extraction of vital information while ignoring irrelevant data. The Decoder is responsible for reconstructing the latent representations generated by the Encoder back into the original time series data. By utilizing LSTMA-AE, we aim to improve the accuracy of anomaly detection, while simultaneously employing mechanistic constraints to mitigate false alarm rates. Experimental results demonstrate that this approach significantly outperforms methods such as polynomial interpolation, random forest, and LSTM-AE in terms of anomaly detection accuracy on field datasets from oilfields, accompanied by a notably lower false alarm rate.

Keywords Water Injection Pump, Multidimensional Time Series Data, Anomaly Detection, Autoencoder

Water flooding is currently an important method for oilfield development. The operational status of injection pumps directly affects the liquid supply capacity of the block, thereby impacting the production of oil wells. If an injection pump malfunctions, oilfield production may be affected. Failure to detect and address pump abnormalities promptly could lead to fires, explosions, and even injuries or damage to personnel and facilities.

In recent years, the research on abnormal detection of injection pumps has been divided into two stages, following the emergence and development of modern detection technologies. The first stage involves fault diagnosis and risk identification of injection pumps based on sensor technology. This is achieved by using various signal processing techniques and selecting sensors like vibration, temperature, and pressure to measure the equipment's operating conditions for monitoring and anomaly detection. Such methods remain widely adopted in modern industrial monitoring due to their reliability and direct applicability. For instance, the use of imaging sensors based on magnetic deposition and flowing dispersion[1], demonstrates how advanced sensor designs can effectively monitor wear debris features, providing valuable insights into mechanical health. Additionally, knowledge graph-based approaches, such as the framework proposed by[2], have shown promise in integrating complex relationships among mechanical components, enabling structured representation of system states for enhanced fault analysis. Building on these methods, many optimization algorithms have been proposed, such as swarm intelligence[3], including particle swarm optimization and its improved versions[4], which has been effectively applied to parameter selection and decision-making in industrial scenarios. However, this method has a high false alarm rate, consuming significant time and effort. The second stage leverages advancements in

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artificial intelligence. Initially, this included risk identification methods based on fault mechanisms, studying the fault mechanisms and post-fault states of injection pumps from a dynamic perspective. Subsequently, methods based on signal analysis emerged, using time-domain, frequency-domain, and time-frequency domain analysis to detect anomalies in injection pumps.

Traditional methods for detecting injection pump anomalies typically rely on specific features, such as polynomial interpolation[5]and k-sigma[6]based on statistical theory, or K-means[7], isolation forests[8], and LOF[9] based on machine learning. These methods generally focus on whether the values of univariate time series data points exceed the corresponding threshold constraints, which overlooks the modeling of inter-dimensional correlations in multivariate time series data and the assessment of anomalies in data trends. Therefore, developing efficient and accurate methods for detecting injection pump anomalies in multivariate time series data remains a challenging task.

With the rapid development of deep learning technology, more researchers are applying it to the processing of multivariate time series data from injection pumps. This is mainly because deep learning models can learn the complex structures of injection pump data through multi-layer nonlinear transformations, better adapting to the characteristics of multivariate time series data and automatically extracting useful features from complex data, something traditional methods struggle to achieve.

Currently, autoencoders (AE) and their variants[10,11] have been extensively studied and applied to practical anomaly detection tasks. An AE is a neural network architecture that performs encoding and decoding operations on multivariate time series data, compressing the input data into latent space representations, and reconstructing the output using these representations. In anomaly detection, they can learn the data distribution observed under normal conditions and classify data that do not conform to this distribution as anomalies using static thresholds.

LSTM-AE[12] is a deep learning model combining LSTM and AE technologies, particularly suited for processing and analyzing time series data. LSTM-AE consists of two main parts: the Encoder and the Decoder. The Encoder captures the temporal dependencies and features in the input sequence through multiple LSTM layers, while the Decoder reconstructs the original time series data from the hidden representation generated by the Encoder, preserving the primary features and information of the input data. However, due to the complexity and diversity of multivariate time series data generated during the operation of injection pumps, with each dimension potentially reflecting different physical quantities such as pressure, flow rate, and temperature, traditional LSTM-AE may struggle to capture complex temporal dependencies and interactions and may not highlight unread information at specific time steps.

To address these challenges, we introduced an attention mechanism into the LSTM-AE model and proposed an attention-based Long Short-Term Memory Autoencoder (LSTMA-AE) to model multivariate time series data from injection pumps comprehensively. Specifically, the LSTMA-AE-based anomaly detection method for injection pumps involves four steps. First, the multivariate time series data of the injection pump is mapped to a high-dimensional space using an Encoder composed of multiple LSTM layers. Second, the attention mechanism captures the temporal dependencies and interactions in the high-dimensional space to fully model the data. Third, a Decoder is used to reconstruct the features in the high-dimensional space back into the original time series. Finally, the reconstruction loss is calculated and compared to a static threshold to determine whether the injection pump time series is anomalous.

Experimental results on the injection pump dataset in oilfields show that, compared to existing time series anomaly detection methods, LSTMA-AE effectively improves the accuracy of injection pump anomaly detection. However, LSTMA-AE tends to misclassify time series with significant operational changes as anomalies. We incorporated engineering experience[13,14]into the model to help it identify and exclude signals caused by normal system variations or significant operational fluctuations rather than true anomalies. To reduce the false alarm rate in the final detection results, we introduced mechanism constraints. Additionally, inspired by the multi-objective trade-off approach proposed by[15], this study applies similar techniques to optimize hyperparameters, focusing on balancing detection accuracy and false alarm rate during the training and validation process.

Related work

Traditional injection pump anomaly detection methods

In the field of injection pump anomaly detection technology[16], proposed a method to calculate pump flow using electrical signals. This method relies on the vibration characteristic curve of the injection pump for the calculation. However, if the actual operating point falls outside the performance range provided by the parameter data, the flow calculation cannot yield accurate values. Moreover, when the vibration characteristic curve of the pump is relatively flat, the calculation error is significant[17]. monitored the entire injection pump system as a whole by observing electrical signals and achieved good results. By testing and analyzing the instantaneous input electrical power of the motor, they could determine various information about the injection pump system[18]. developed an enhanced random forest method to detect operational anomalies in injection pumps within industrial processes[19]. proposed an anomaly detection method for injection pumps based on wavelet singularity and BP neural networks in the context of controlled fault interruption. This method involves sampling the three-phase voltage and current of a model based on a short-circuit fault simulation model, followed by a transformation using DB4 wavelets. After extracting the absolute value of the modulus maximum of wavelet transform coefficients at singular points, these values are used as input feature vectors for a BP neural network[20]. introduced an adaptive genetic algorithm to optimize BP networks for improving fault recognition accuracy in the context of injection pump faults. Similarly, other variants of RNN, such as the Gated Recurrent Unit (GRU), have demonstrated their effectiveness in sequential data tasks. For instance[21], utilized a hierarchical framework combining CNN and GRU for structural damage detection, showcasing the adaptability of GRU in capturing both spatial and temporal features, which complements LSTM's capabilities in long sequence modeling.

These methods typically utilize univariate time series for fault detection in injection pumps, focusing on whether the values at individual time points exceed the corresponding threshold constraints. This approach overlooks the modeling of correlations between different dimensions in multivariate time series data and the measurement of anomalies in data trends. Although certain methods, such as the use of quadratic regression for trendline analysis in[22], are capable of capturing trends in multivariate data, they often struggle with the complexity of inter-variable relationships and non-linear dependencies. Similarly, image transformation techniques, such as those employed for partial discharge source identification in high-speed train cable terminals[23], have shown promise in feature extraction and visualization but are limited in addressing the temporal dynamics of multivariate datasets. In recent years, the application of transformer-based architectures has brought deep learning methods into focus for fault detection tasks. Models leveraging self-attention mechanisms, such as those proposed by[24], have demonstrated significant potential in capturing complex relationships in multidimensional data. In this paper, we use multivariate time series for anomaly detection in injection pumps, fully leveraging the relationships between different dimensions of the time series data.

Deep learning-based anomaly detection methods

In the context of unsupervised frameworks[25], studied the problem of anomaly detection and proposed a method that uses LSTM neural networks for clustering time series data. They detect anomalies by comparing the input data sequence with the output of the clustering model. Similarly[26], introduced an LSTM-based anomaly detection model, which is trained on normal data and uses reconstruction error as an anomaly score, detecting anomalies by comparing this score to a threshold[27]. proposed a denoising autoencoder model that can restore noisy input data to its noise-free state. Compared to a regular AE, the increased difficulty in reconstruction enables the encoder to learn more representative latent features[28]. proposed an anomaly detection method based on the Variational AutoEncoder (VAE), a probabilistic model that uses variational inference to determine anomalies through reconstruction probability. Additionally, Sparse Autoencoders (SAE)[29]have been applied to time series anomaly detection by adding sparsity constraints to the hidden layer nodes, encouraging sparse feature learning, which forces the neural network to learn more representative features from the training data[30]. introduced a method using two autoencoders (a traditional autoencoder and a sparse autoencoder) in tandem for anomaly detection. This approach has a strong complementary effect, allowing the second autoencoder to detect anomalies that the first might have missed[31]. proposed a relational autoencoder model that extracts high-level features from the data itself and its internal relationships, reducing reconstruction errors[32]. developed a deep autoencoding Gaussian mixture model (DAGMM) for unsupervised anomaly detection. This model generates low-dimensional representations and reconstruction errors for each input data point using a deep autoencoder, which are then fed into a Gaussian Mixture Model (GMM) to evaluate the likelihood of each data point. The higher the likelihood, the greater the probability of an anomaly.

Inspired by the[33,34], our model leverages attention mechanisms to capture temporal dependencies and spatial relationships in multivariate time series data. We enhance the AutoEncoder by incorporating an attention mechanism module, which models the features of each dimension in a high-dimensional space. Additionally, we introduce mechanism constraints to improve anomaly detection accuracy while reducing false alarm rates, making LSTMA-AE more suitable for injection pump anomaly detection.

Method

To address the issue of detecting operational anomalies in multivariate time series data from injection pumps, improving detection accuracy while reducing false alarm rates, this paper proposes a multivariate time series injection pump anomaly detection method based on LSTMA-AE and mechanism constraints. The process flow is illustrated in Figure 1. First, we use LSTMA-AE to reconstruct the input time series; the reconstruction loss is then calculated. If the reconstruction loss exceeds a predefined threshold, the time series is classified

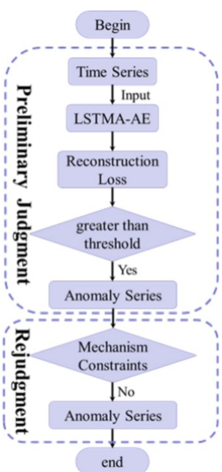

Fig. 1. The process flow of anomaly detection.

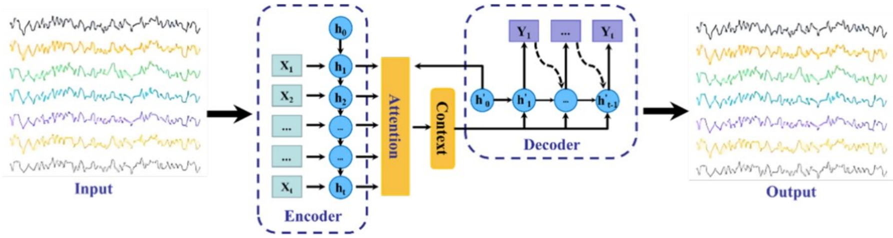

Fig. 2. The structure of LSTMA-AE.

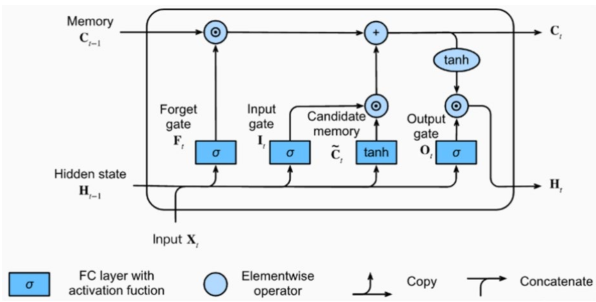

Fig. 3. The structure of LSTM.

as anomalous; otherwise, it is deemed normal. Next, mechanism constraints are applied to the time series classified as anomalous. If the series does not meet the conditions specified by the mechanism constraints, it is confirmed as an anomaly; otherwise, it is classified as normal. In this section, subsections 3.1 and 3.2 will focus on describing LSTMA-AE and the mechanism constraints for injection pump anomaly detection.

LSTMA-AE

This section introduces the proposed LSTMA-AE. The LSTMA-AE model primarily consists of four components: the Time Feature Extraction Module (Encoder), the Attention Module (Attention Layer), the Data Reconstruction Module (Decoder), and the Autoencoder Framework. The overall framework is illustrated in Figure 2.

LSTM is a special type of Recurrent Neural Network (RNN) architecture[35]. The core innovation of LSTM lies in its introduction of three gating units: the input gate, forget gate, and output gate. These gating units allow LSTM to adaptively decide which information should be retained in the memory cell, thereby optimizing its performance in modeling long sequences.

LSTM-AE

LSTM-AE is a deep learning model that combines LSTM and AE technologies, particularly suited for processing and analyzing time series data. The LSTM-AE model consists of two main components: the Encoder and the Decoder. The Encoder captures the temporal dependencies and features of the input sequence through multiple LSTM layers, while the Decoder is responsible for reconstructing the original time series data from the hidden representations generated by the Encoder, preserving the main features and information of the input data. Figure 3 illustrates the basic structure of LSTM. In the figure, $X_t$represents the input at the current time *t*. H_t denotes the hidden state at the current time *t*. F_t is the forget gate at the current time *t*. I_t is the input gate at the current time *t*. O_t is the output gate at the current time *t*. $\tilde{C}_t$ is the candidate memory cell at the current time *t* and C_t is the memory cell at the current time *t*. The specific calculation is shown as equation 1,

$$\begin{aligned} F_t &= \sigma\left(W_f \cdot [H_{t-1}, X_t] + b_f\right) \\ I_t &= \sigma\left(W_i \cdot [H_{t-1}, X_t] + b_i\right) \\ \tilde{C}_t &= \tanh\left(W_C \cdot [H_{t-1}, X_t] + b_C\right) \\ C_t &= F_t^* C_{t-1} + I_t^* \tilde{C}_t \\ o_t &= \sigma\left(W_o \cdot [H_{t-1}, X_t] + b_o\right) \\ H_t &= O_t^* \tanh\left(C_t\right) \end{aligned} \tag{1}$$

where σ represents the Sigmoid activation function, and *tanh* denotes the hyperbolic tangent activation function. W_i, W_f, W_o and $W_c$represent the weight matrices for the input, forget, output gates, and candidate memory cell state, respectively. b_i, b_f, b_o and b_c denote the bias terms for the input, forget, output gates, and candidate memory cell state, respectively. $\odot$ represents the element-wise multiplication operator.

Attention

Due to the complexity and diversity of the multivariate time series data generated during the operation of injection pumps, where each dimension may reflect different physical quantities such as pressure, flow rate, and temperature, traditional LSTM-AE models may struggle with capturing complex temporal dependencies and interactions. They may also have difficulty highlighting important information across different time steps or specific unread information. Therefore, we have introduced an attention mechanism into the LSTM-AE model. This paper incorporates a temporal pattern attention mechanism into the calculation of the hidden states in the LSTM-AE model to enhance the model's ability to learn from information at different time steps. The attention mechanism helps the model dynamically adjust the contribution of each time step's input information to the hidden state, better capturing important information while ignoring irrelevant details, thereby improving the model's modeling capability. The computation flow is shown in Figure 4. For each decoder time step *t*, we calculate the attention scores between all the encoder's hidden states $h_i (i = 1, 2, ..., t)$ and the current hidden state h'_t of the decoder (equation 2).

$$e_{ti} = \text{score}\left(h'_t, h_i\right) \tag{2}$$

Here, we use the additive attention mechanism, with its calculation given by equation 3.

$$e_{ti} = v^T \tanh\left(W_1 h'_t + W_2 h_i\right) \tag{3}$$

In this context, v^T, W_1, W_2 are learnable parameter matrix. The attention scores are then normalized to obtain the attention weights for each hidden state (equation 4).

$$\alpha_{ti} = \frac{\exp\left(e_{ti}\right)}{\sum_{j=1}^{t} \exp\left(e_{tj}\right)} \tag{4}$$

Next, the weighted average context vector is calculated as equation 5.

$$c_t = \sum_{i=1}^{t} \alpha_{ti} h_i \tag{5}$$

The context vector C_t contains the weighted information of all the encoder's hidden states, which better represents the important feature information of the input sequence at the current time step. The context vector C_t is then introduced into the decoder. The LSTM unit of the decoder is calculated as equation 6,

$$h'_t = \text{LSTM}(\left[y_{t-1}, c_t\right], h'_{t-1}) \tag{6}$$

where y_{t-1} represents the previous step's decoder output, c_t represents the current time step's context vector, and h'_{t-1} represents the previous step's hidden state. The current hidden state h'_t is then mapped to the output space through a linear transformation to obtain the current time step's output y_t (equation 7).

$$y_t = \text{ Linear } \left(h'_t\right) \tag{7}$$

In this way, when the decoder generates a new hidden state h_t at each step, it fully utilizes the global information from the encoder's hidden states, thereby improving the model's performance. By introducing the attention mechanism, our LSTMA-AE model demonstrates superior performance in the anomaly detection of injection pumps using multivariate time series data. It can better capture and balance information across different time steps, effectively seizing complex temporal dependencies and interaction effects. The model performs well on training data and also exhibits strong generalization ability on unseen data, making it more robust in practical applications.

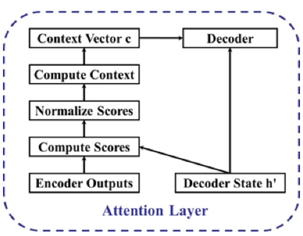

Fig. 4. The computation flow of Attention.

Mechanism constraints

Anomaly detection methods are typically used to identify data points in a dataset that deviate significantly from expected patterns. However, in real-world applications, some seemingly anomalous points do not warrant alarms, such as fluctuations caused by monitoring instrumentation faults or operational changes. These data points are unlikely to threaten system safety or efficiency and can be effectively filtered using mechanistic constraint rules. To enhance the accuracy and effectiveness of the anomaly detection method, we propose three specific scenarios as mechanism constraints:

- **Sudden short-term spikes in electrical parameters with rapid recovery:** These are typically caused by transient electrical fluctuations or events like equipment start-ups or shutdowns. Although such spikes deviate from normal patterns, their quick recovery minimizes any long-term impact. A threshold is defined to classify spikes that return to normal within a specified time window as normal variations.
- **Minor short-term fluctuations in outlet pressure at low flow and current levels:** When flow and current are low, outlet pressure may exhibit small fluctuations due to minor pipeline disturbances, sensor sensitivity limits, or measurement noise. These fluctuations are considered normal dynamic responses of the system. By analyzing flow and current trends, small pressure variations under such conditions are excluded from anomalies.
- **Short-term increases in inlet pressure at low flow and current levels:** In certain operating conditions, inlet pressure may temporarily rise due to localized flow changes or environmental factors like temperature and humidity. A reasonable pressure fluctuation threshold, combined with flow and current levels, is used to filter out such cases as normal operational variations.To ensure accurate identification of these scenarios, the mechanism constraints are integrated into the preprocessing phase of the anomaly detection algorithm through the following steps:
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- **Rule Design:** Each constraint rule is based on real-world operating conditions and data characteristics, specifying thresholds for short-term variations, recovery windows, and flow/current levels to accommodate diverse scenarios.
- **Data Preprocessing:** Before entering the anomaly detection system, data is checked against the mechanism constraints. Data points meeting the predefined rules are flagged as normal fluctuations and excluded from the anomaly detection sequence.By embedding these mechanism constraints, the proposed method effectively reduces false alarms caused by normal fluctuations while maintaining high sensitivity and accuracy for true anomalies. This approach ensures reliable performance in practical applications.

Experiment

Data description

In this study, we utilized an injection pump operation dataset collected from Shengli Oilfield to conduct research on detecting anomalies in injection pump operations. This dataset includes sensor readings and other relevant variables recorded at multiple time points, providing detailed information on the equipment's operation under different working conditions. To ensure the effectiveness of the model and the fairness of the evaluation, the dataset was divided into training and test sets in a 7:3 ratio. The training set was used to train the model, while the test set was used to evaluate its performance. Notably, all anomalies within the time series in the dataset have been clearly labeled, providing reliable label information for subsequent supervised learning.

Given the significant differences in magnitude across the time series attributes within the same dataset, we standardized the data samples to ensure that they follow a standard normal distribution with a mean of 0 and a variance of 1. Additionally, to enhance the model's generalization capability, each time series sample used in training was randomly flipped with a probability of 50%.

To further validate the generalization capability of the proposed model, we extended our experiments to include the FordA dataset from the UCR archive and the Server Machine Dataset (SMD)[36]. The FordA dataset originates from a competition held during the 2008 IEEE World Congress on Computational Intelligence, designed to diagnose the presence of a specific mechanical symptom in an automotive subsystem. Each sample consists of 500 sequential measurements of engine signals. The SMD dataset, on the other hand, provides five weeks of monitoring data from a leading Internet company, containing readings from 38 distinct sensors deployed across 28 server machines. This dataset offers a comprehensive benchmark for time-series anomaly detection due to its real-world complexity and diverse sensor readings.The anomaly proportions of the UCR, SMD and our datasets are shown in the Table 1.

Experimental design

The model was implemented using the PyTorch deep learning framework and trained on an Nvidia Tesla P100 16G GPU. The Adam optimizer with a learning rate of 0.01 was used, and the training process involved

Datasets	Proportion of anomalies
UCR	0.515
SMD	0.042
OURS	0.107

Table 1. Proportion of anomalies in datasets.

Method	Recall	Precision	F1
EWMA	0.510	0.490	0.500
PI	0.490	0.510	0.500
IF	0.610	0.490	0.543
LOF	0.540	0.490	0.513
LSTM	0.821	0.833	0.827
CBAM-AE	0.846	0.876	0.860
RESNET-AE	0.869	0.798	0.831
LSTMA-AE	0.876	0.883	0.879

Table 2. Comparative experimental results of different methods on UCR dataset.

Method	Recall	Precision	F1
EWMA	0.480	0.500	0.490
PI	0.470	0.500	0.485
IF	0.510	0.520	0.515
LOF	0.480	0.500	0.490
LSTM	0.837	0.812	0.824
CBAM-AE	0.837	0.864	0.850
RESNET-AE	0.821	0.834	0.827
LSTMA-AE	0.846	0.870	0.858

Table 3. Comparative experimental results of different methods on SMD dataset.

Method	Recall	Precision	F1
EWMA	0.412	0.425	0.418
PI	0.432	0.416	0.424
IF	0.419	0.421	0.420
LOF	0.422	0.457	0.439
LSTM	0.837	0.830	0.833
CBAM-AE	0.813	0.821	0.817
RESNET-AE	0.833	0.843	0.838
LSTMA-AE	0.868	0.869	0.868
LSTMA-AE+Mechanism Constraints	**0.926**	**0.915**	**0.920**

Table 4. Comparative experimental results of different methods on our dataset.

1000 iterations. To validate the effectiveness of the proposed LSTMA-AE and mechanistic constraint-based anomaly detection method for injection pump operation, a comparative experiment was conducted with several contemporary time series anomaly detection methods, including Exponential Weighted Moving Average (EWMA), polynomial interpolation (PI), isolation forest (IF), Local Outlier Factor (LOF), ResNet-AE[37], and CBAMA-AE[38], an anomaly detection algorithm specifically designed for oilfield pumping unit operations. The evaluation metrics included recall, precision, and F1 score. Recall measures the proportion of true positive anomalies correctly identified by the model, precision quantifies the proportion of detected anomalies that are actually true anomalies, and F1 score is the harmonic mean of precision and recall, providing a balanced measure of the model's performance. Additionally, an ablation study was performed to assess the impact of each module on the effectiveness of time series anomaly detection in injection pump and oilfield pumping unit operations.

Experimental result

We conducted comparative experiments on three datasets respectively. For details, please refer to Table 2, 3, 4. During thet in-depth analysis of the experimental results, we observed that the first four detection methods demonstrated high efficiency in identifying significant fluctuations in key operational parameters of the injection pump (as shown in Figure 5). This characteristic ensures their effectiveness in detecting major anomalies. However, these methods exhibit a marked decrease in sensitivity when faced with subtle and gradual changes in operational parameters, leading to a lower recall rate for anomaly detection. Consequently, they failed to adequately capture all the minor variations that should have been flagged as anomalies.

In contrast, the LSTM-AE model, with its unique structural advantages, displayed a high sensitivity to minor changes in operational parameters. This attribute is directly reflected in its higher recall rate, demonstrating the

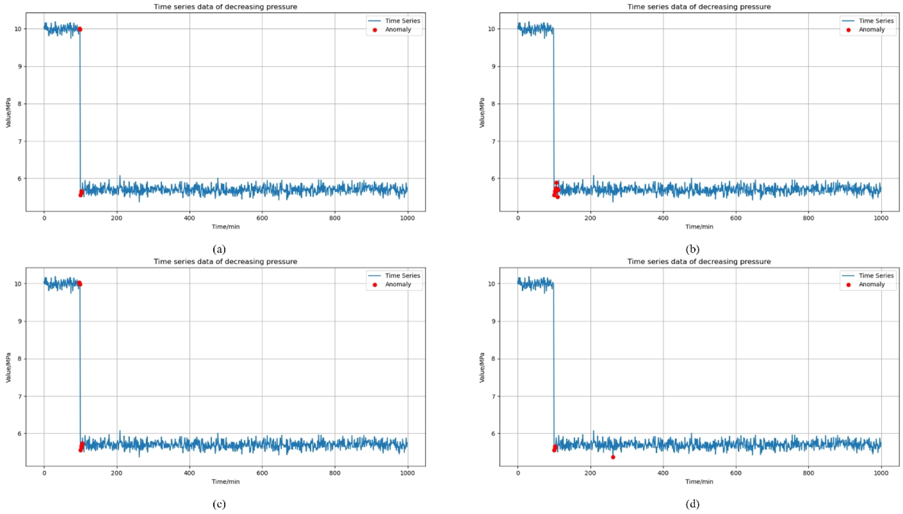

Fig. 5. Abnormal detection results of the four methods. (**a**) EWMA. (**b**) PI. (**c**) IF. (**d**) LOF.

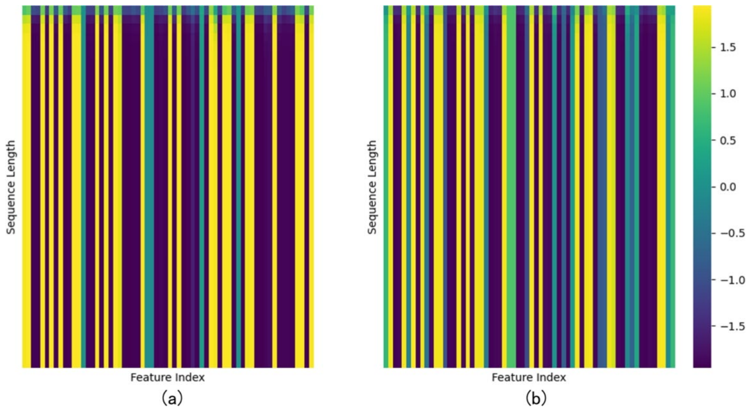

Fig. 6. (**a**) Heatmap of Original Decoder Output. (**b**) Heatmap of Attention-based Decoder Output.

model's superior ability to detect early signs of anomalies. Notably, by incorporating an attention mechanism into the LSTM-AE model (resulting in the LSTMA-AE model), we further enhanced the model's ability to perceive subtle dynamic changes in operational parameters. This not only reinforced its recall rate advantage but also suggested greater potential for precise and nuanced anomaly detection.

The impact of the attention mechanism can be visually examined through the attention heatmaps, as shown in Figure 6. These heatmaps illustrate how the LSTMA-AE model allocates attention across different feature dimensions over the sequence length. The left side of Figure 6 represents the attention heatmap of the standard LSTM-AE model without the attention mechanism, where the focus is relatively uniform and lacks explicit discrimination of subtle parameter changes. In contrast, the right side shows the attention heatmap of the LSTMA-AE model with the attention mechanism, which demonstrates a more focused and dynamic allocation of attention.

This focused attention allows the model to prioritize features that are indicative of early anomaly onset, thereby achieving enhanced recall rates. Moreover, the attention heatmap provides interpretability by highlighting the specific features and time steps that contribute significantly to the anomaly detection process.

The reconstructed data images using LSTMA-AE, as shown in the Figure 7, demonstrate the identification accuracy of our experimental results in terms of long-term abnormal trends and short-term fluctuations.

However, the LSTMA-AE model showed some vulnerability to noise interference, which is inevitable in complex industrial environments. This was evidenced by a decrease in its precision metric, which somewhat limits its broad applicability and reliability in real-world scenarios. To address this challenge, we innovatively introduced mechanistic constraints to optimize the LSTMA-AE model. This strategy, grounded in physical principles and the inherent logic of system operations, effectively filtered out irrelevant noise, significantly improving the model's noise resistance and overall precision. This improvement not only solidified LSTMA-AE's leading position in anomaly detection but also provided a robust theoretical foundation and practical support for its application in complex industrial environments.

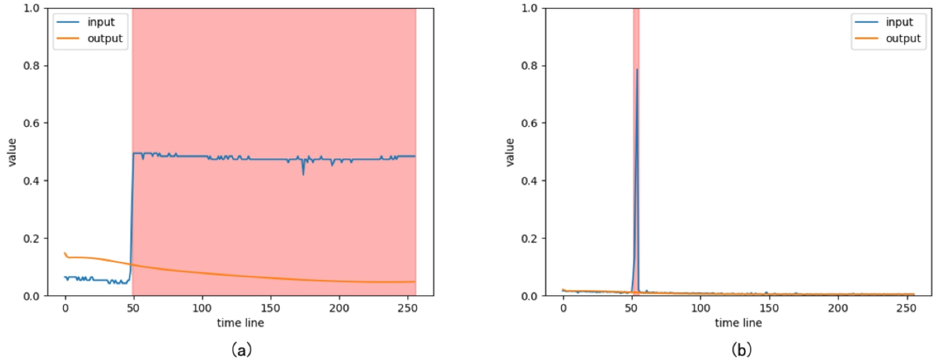

Fig. 7. (**a**) Long-term anomalous data. (**b**) Short-term anomalous data.

In summary, through systematic experimental analysis, this study not only revealed the limitations of traditional detection methods in detecting subtle anomalies but also validated the significant effectiveness of LSTM-AE and its enhanced version (LSTMA-AE with mechanistic constraints) in improving anomaly detection sensitivity, precision, and noise resistance. This research offers valuable insights and references for intelligent maintenance in injection pumps and broader industrial applications.

Conclusion

This paper delves deeply into a multidimensional time series analysis method based on Long Short-Term Memory Autoencoder (LSTMA-AE) combined with mechanistic constraints, which is applied to anomaly detection in injection pump operation. This approach artfully merges the nonlinear feature extraction capabilities of deep learning with domain-specific mechanistic knowledge constraints, thereby constructing an efficient and robust anomaly detection framework.

Specifically, the introduction of LSTMA-AE enables the effective capture of temporal dependencies and complex dynamic characteristics within the operational data of injection pumps. This notably bolsters the model's capacity to represent normal operational modes. Meanwhile, by integrating mechanistic constraints, the model's comprehension of the inherent rules governing physical processes is enhanced. This further narrows the anomaly detection space, effectively reduces the false alarm rate, and guarantees the accuracy and reliability of the detection outcomes. Experimental results demonstrate that this method showcases remarkable performance in identifying anomalous data during injection pump operation. It not only enhances the precision of anomaly detection but also curtails unnecessary shutdown inspections, consequently optimizing maintenance efficiency.

In practical terms, the proposed multidimensional time series anomaly detection method based on LSTMA-AE and mechanistic constraints offers substantial technical support and theoretical reference for formulating industrial equipment health management and maintenance strategies. It can be widely applied in various industrial settings where injection pumps or similar equipment are in use, helping enterprises to better monitor equipment conditions, reduce maintenance costs, and avoid production disruptions caused by unexpected breakdowns.

Looking ahead, there are several promising directions for future work. For instance, researchers could explore adapting this method to other types of industrial machinery with different operational characteristics and time series data patterns, expanding its scope of application. Additionally, further investigations could be conducted to refine the combination of deep learning models and mechanistic constraints, perhaps by incorporating more advanced forms of domain knowledge or optimizing the way these constraints are imposed on the model. There is also potential to develop more comprehensive evaluation metrics to better assess the performance of the method in different real-world scenarios. Overall, this method holds significant academic value and is expected to inspire more research efforts in the intersection of deep learning and industrial equipment anomaly detection, thus opening up a broader range of possibilities for improving industrial equipment reliability and maintenance practices.

Data availability

We understand that data sharing is crucial for advancing research in industrial anomaly detection. However, due to the proprietary nature of the private dataset used in this study, it involves industrial secrets and therefore cannot be shared publicly. We are, however, happy to address any questions or provide further clarifications related to the content of this paper. The **SMD dataset** is a publicly available benchmark for anomaly detection and can be accessed at https://github.com/NetManAIOps/OmniAnomaly. The **UCR Time Series Archive** is widely used in time-series research and can be downloaded from https://www.cs.ucr.edu/~eamonn/time_series_data_2018/.

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Additional information

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